CSE461

Section : 06

Group : 01

Semester : Summer_2025

Robot Project Proposal

Member Details:

1. Name: B M Rauf ID: 22201782
2. Name: Azmari Sultana ID: 22201949
3. Name: Sayeba Nasir ID: 22201354

Submission Date: 18 – 07 – 2025

Instructors: CSDM

Project Idea 1

1. Project Title: Fire Fighting Robocar

2. Purpose

Objective: The primary goal of this project is to develop an autonomous firefighting robot capable of detecting fire using flame sensors and extinguishing it using a water spray mechanism, making it useful in small-scale indoor fire emergencies.

Scope: This project involves developing a firefighting robot capable of operating in small-scale indoor environments such as homes, offices etc. It will use flame sensors to detect the presence of fire and respond by navigating toward the source and activating an onboard water spray mechanism to extinguish it. The design focuses on autonomous movement, environmental awareness, and effective fire suppression without human assistance.

Significance: Indoor fires can pose serious risks if not addressed quickly. This project presents a cost-effective, autonomous robotic solution aimed at early fire detection and control. By using flame sensors and an automatic extinguishing system, the robot minimizes the need for human presence in potentially dangerous situations. It highlights the application of robotics and sensor-based automation in real-world safety scenarios, with potential to reduce fire-related damage and enhance emergency response in confined spaces.

3. Components

- Microcontroller: Arduino UNO R3
- Sensors:
  - Flame Sensor
  - Smoke Sensor
  - Water Level Sensor
- Actuators:
  - DC Motor
  - Servo Motor
  - DC Submersible Pump

- Body/Chassis: A four-wheel robotic car chassis
- Additional Components:
  - Motor Driver Module
  - Battery
  - Wires
  - Switch
  - USB cables

[Add Diagrams or Figures if needed]

4. Cost Breakdown

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	Arduino Uno R3	1	990	990
2	Flame Sensor	5	70	350
3	Smoke Sensor	1	200	200
4	Water Level Sensor	1	50	50
5	DC Motors	2	45	90
6	Servo Motor	1	450	450
7	DC Submersible Pump	1	150	150
8	Others			1000
Total Cost (BDT)				3280

5. Functionality Breakdown

- **Functionality 1: Smoke and Fire Detection**

  ○ **Overview:** The smoke and flame detection sensors will detect the presence of smoke and fire.
  ○ **Working Procedure:** The flame sensors will be positioned in different directions to cover 180 degrees. These sensors will be connected to the Arduino's digital pins. When any sensor detects a flame, it will send a HIGH signal to the Arduino.

- **Functionality 2: Navigation and Movement**

  ○ **Overview:** The robot will move automatically toward the fire sources.

  ○ **Working Procedure:** Motor driver module will be used to control the DC motors. When it receives control signals from the Arduino, it will drive the motors accordingly to navigate toward the fire.

- **Functionality 3: Fire Extinguishing**

  ○ **Overview:** Once a fire is detected, the robot will activate the pump and spray water using a servo-based nozzle.
  ○ **Working Procedure:** When the Arduino receives input from any flame sensor, it will trigger the submersible pump and rotate the servo motor to aim the water pipe toward the fire. The pump will pull water from the container and spray it out via the pipe to extinguish the fire.

6. Potential Challenges

- **Technical Challenges:** Accuracy of flame sensors in bright lighting conditions, Servo motor precision in aiming water, Limited water capacity in the container etc.
- **Design Challenges:** Space constraints, Weight distribution etc.
- **Integration Challenges:** Sensor Calibration, Water System Integration, Wiring Complexity etc.

Project Idea 2

1. Project Title: Autonomous Cat Feeder Robot

2. Purpose

Objective: The primary goal of this project is to design an autonomous robot that can feed a pet cat at scheduled times without human intervention. The robot will be capable of storing dry or wet cat food and automatically dispensing it when needed, ensuring your cat receives its meals on time even if the owner is away.

Scope: This robot will serve as a fully autonomous cat feeder, capable of adjusting feeding portions, detecting when the food container is empty, and replenishing food if needed. It will be designed for use in homes or apartments, where pet owners may not always be available to feed their cat at regular intervals.

Significance: The autonomous cat feeder robot automates an essential pet care task, giving owners more flexibility and ensuring their cats are fed at the appropriate times. It helps eliminate concerns about missed feeding times due to busy schedules, travel, or forgetfulness.

3. Components

- Microcontroller: Arduino Uno R3
- Sensors:
  - Weight Sensor
  - Ultrasonic Sensor
  - IR Sensor
- Actuators:
  - Servo Motor
  - DC Motor
  - Stepper Motor

- Body/Chassis: A stable frame to hold the food container, motors, and electronic components, while ensuring it is safe for a pet's interaction.
- Additional Components:
    - Motor Driver Module
    - Relay Module
    - Battery
    - Wires
    - Switch
    - USB cables
    - Food Container

[Add Diagrams or Figures if needed]

4. Cost Breakdown

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	Arduino Uno R3	1	990	990
2	Weight Sensor	1	1	380
3	Ultrasonic Sensor	1	100	100
4	IR Sensor	1	70	70
5	DC Motors	2	45	90
6	Servo Motor	1	450	450
7	Stepper Motor	1	250	250
8	Others			1000

Total Cost (BDT)	3330

5. Functionality Breakdown

- **Functionality 1: Automatic Food Dispensing**

  ○ **Overview:** Automatically dispense a portion of food at designated times.

  ○ **Working Procedure:** A weight sensor will monitor the amount of food dispensed into the bowl to ensure the correct portion is delivered. Actuators will operate the food dispensing mechanism with precision, releasing the desired quantity.

- **Functionality 2: Low Food Detection**

  ○ **Overview:** Automatically detect when the food container is low or empty.

  ○ **Working Procedure:** The ultrasonic sensor will monitor the food level in the container, and if it drops below a set threshold, the robot will trigger a low-food alert via buzzer.

- **Functionality 3: Bowl Monitoring and Feeding Confirmation**

  ○ **Overview:** Ensure the food bowl is present and confirm feeding has occurred.

  ○ **Working Procedure:** IR sensor will be placed to detect whether the bowl is in place before dispensing begins. After dispensing, the weight sensor confirms that food was successfully dropped into the bowl. If no food is detected, the system retries or triggers an alert.

6. Potential Challenges

- **Technical Challenges:** Sensor Accuracy, Portion Control etc.
- **Design Challenges:** Size of the Robot, Stability, Durability etc.
- **Integration Challenges:** Sensor Calibration, Motor Precision, Power supply regulation etc.

Project Idea 3

1. Project Title: Smart Autonomous Vacuum & Wet Floor Mopping Robot

2. Purpose

Objective: The primary goal of this project will be to design and develop a smart autonomous vacuum and wet floor cleaning robot using Arduino. The robot will be capable of navigating its environment, detecting and avoiding obstacles and wet surfaces, and simulating both vacuum suction and wet floor cleaning. It will also respond to sound-based commands, such as claps or other distinct noises, making it interactive and responsive to its surroundings.

Scope: The robot will be designed for use in small indoor environments such as homes, classrooms, or offices. It will perform basic autonomous cleaning by navigating the space intelligently, detecting and avoiding obstacles, and simulating both vacuuming and wet floor mopping. Additionally, since it responds to user-triggered sound inputs, such as claps to start or stop cleaning operations, it will be interactive and user-friendly.

Significance: This project will demonstrate how embedded systems and sensor integration can be used to develop a cost-effective alternative to commercial robotic vacuum cleaners. By incorporating wet-floor detection and sound-triggered control, the robot will offer enhanced interactivity and functionality. The project will also serve as a practical example of applying robotics and automation principles in real-world scenarios, promoting hands-on learning and innovation in smart home technology.

3. Components

- Microcontroller: Arduino Uno R3
- Sensors:
  - Ultrasonic Sensor
  - Sound Sensor
  - Water Detection Sensor
  - IR Obstacle Avoidance Sensor

- Actuators:
  - Servo Motor
  - DC Motor
  - Gear Motors with wheels

- Body/Chassis: Cardboard chassis
- Additional Components:
  - Motor Driver Module
  - Battery
  - Wires

[Add Diagrams or Figures if needed]

4. Cost Breakdown

No	Components	Quantity	Unit Cost (BDT)	Total Cost (BDT)
1	Arduino Uno R3	1	990	990
2	Ultrasonic Sensor	1	100	100
3	Sound Sensor	1	73	73
4	Servo Motor	1	450	450
5	DC Motor	1	45	45
6	Others		2000	2000
Total Cost (BDT)				3658

5. Functionality Breakdown

- **Functionality 1: Vacuum and Mopping**

  ○ **Overview:** Detects the type of surface and activates either vacuuming or mopping, depending on whether the floor is dry or wet.
  ○ **Working Procedure:** A moisture sensor will be used to detect the surface type. If the floor is dry, the robot will vacuum the surface; if the surface is wet, the robot will mop it.

- **Functionality 2: Obstacle Avoidance**

  ○ **Overview:** Detect and avoid obstacles using an ultrasonic sensor to prevent collisions and ensure smooth navigation.
  ○ **Working Procedure:** An ultrasonic sensor will scan the environment. If an object is detected within a set distance, the robot will stop, change its direction, and resume cleaning along a new path.

- **Functionality 3: Sound Triggered Control**

  ○ **Overview:** Respond to sounds like claps to start or stop cleaning.

  ○ **Working Procedure:** A sound sensor will detect specific sounds, such as claps. When a valid sound event is detected, the robot will toggle between cleaning mode and idle mode.

6. Potential Challenges

- **Technical Challenges:** Sensor accuracy, Motor Control, Power Management etc.
- **Design Challenges:** Surface Compatibility, Components Placement etc.
- **Integration Challenges:** Component Compatibility, Wiring Complexity etc.